

CS 606: Computational Modeling of Social Systems

Course Code	CS 606
Course Name	Computational Modeling of Social Systems
L-T-P-C	3-0-0-3
Prerequisites	IC 110 Engineering Mathematics and IC 150 Computation for Engineers; or, after instructors permission
Intended for	3rd, 4th year B. Tech./M.S./Ph.D.
Distribution	Discipline-elective for CSE B. Tech. students; Free elective for other B. Tech. students
Approval	4th Senate (May 2013)

Course Contents

- **Learning in Complex Systems:** Policy Resistance, Counterintuitive Behavior of Social Systems; Learning and feedback process; Barriers in Learning; Requirements for Learning in Complex Systems; Virtues and Pitfalls of Virtual Worlds; Importance of simulation. [4 Lectures]
- **The Modeling Process:** Purpose of Modeling; Steps in the Modeling Process; Problem Articulation; Formulating a Dynamic Hypothesis; Formulating a Simulation Model; Testing; Policy Design and Evaluation; Causal Loop Diagrams; Fundamental Modes of Dynamic Behavior; Dynamics of Stocks and Flows: First-order Systems, Positive Feedback and Exponential Growth, Multiple-Loop Systems, Non-linear First-Order Systems. [8 Lectures]
- **Applications of Modeling Process:** Modeling S-Shaped Growth; Modeling Epidemics (e.g., Modeling the HIV/AIDS Epidemic); Modeling Innovation Diffusion: Modeling New Ideas and New Products (The Bass Diffusion Model). [6 Lectures]
- **Feedback Delays:** Duration and Dynamics of Delays; Defining Delays; Material Delays: Structure and Behavior; Information Delays: Structure and Behavior; Response to Variable Delay Times; Estimating the Duration and Distribution of Delays; Examples: Forecasting Semiconductor Demand. [8 Lectures]
- **Modeling Decision Making:** Principles for Modeling Decision Making; Formulating Rate Equations; Common Pitfalls; Human Decision Making: Bounded Rationality or Rational Expectations; Cognitive Limitations; Individual and Organizational Responses to Bounded Rationality; Intended Rationality; Case Study: Modeling High-Tech Growth Firms. [8 Lectures]
- **Validation and Model Testing:** Difficulties in Validation and Verification; Questions Model Users Should Ask: Types of Data, Documentation, Replicability, Protective versus Reflective Modeling; Model Testing in Practice (Boundary Adequacy Tests, Structure Assessment Tests, Dimensional Consistency, Parameter Assessment, Extreme Condition Tests). [8 Lectures]

Text Books

1. John D Sterman, **Business Dynamics: Systems Thinking and Modeling for a Complex World**, McGraw-Hill, 2000

References

1. Donella H. Meadows, **Thinking in Systems: A Primer** (Paperback), Chelsea Green Publishing, 2008.
2. John Morecroft, **Strategic Modelling and Business Dynamics: A Feedback Systems Approach**, John Wiley & Sons, 2007.